05. 2011 Machine Learning Dataset Preparation for Temporal Comparative Analysis

Section-1. Objective

The objective of this notebook is to prepare a machine learning-ready dataset from the 2011 Toronto Neighbourhood Profiles for temporal comparative analysis.

The notebook identifies and extracts socioeconomic indicators that are comparable with the 2016 machine learning-ready dataset, transforms the original profile-style dataset into a neighbourhood-level analytical dataset, performs data quality assessment, and exports the cleaned dataset for use in Notebook 5.

Input

- `neighbourhood-profiles-2011-140-model.csv`

Output

- `neighbourhood_profiles_2011_ml_ready.csv`

Main Tasks

- Load the raw 2011 Toronto Neighbourhood Profiles dataset.
- Validate the dataset structure and quality.
- Identify and extract the selected socioeconomic indicators.
- Construct a machine learning-ready neighbourhood-level dataset.
- Perform data quality assessment.
- Export the processed dataset for temporal comparative analysis.

In [1]:
```
# Import libraries

import pandas as pd
import numpy as np

pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", None)
```

In [3]:
```
# Load the raw dataset

raw_df = pd.read_csv(
    "../../data/raw/neighbourhood-profiles-2011-140-model.csv")

print("Dataset Shape:", raw_df.shape)

display(raw_df.head())
```

```
Dataset Shape: (1537, 145)
```

	_id	Category	Topic	Attribute	City of Toronto	Agincourt North	Agincourt South-Malvern West	Alderwood	An
0	1	Population	Population	Population, 2011	2615060.0	30279.0	21988.0	11904.0	2917
1	2	Population	Population	Population, 2006	2503281.0	30156.0	21562.0	11656.0	2748
2	3	Population	Population	Population percentage change, 2006 to 2011	4.5	NaN	NaN	NaN	N
3	4	Population	Population	Population density per square kilometre	4149.5	NaN	NaN	NaN	N
4	5	Population	Dwellings	Total private dwellings	1107851.0	9341.0	7861.0	4840.0	1717

Section-2 Data overview

Inspect the structure of the raw 2011 Toronto Neighbourhood Profiles dataset by examining its dimensions, column names, data types, and missing values. This assessment provides an overview of the dataset and identifies any structural characteristics that should be considered during feature extraction and preprocessing.

In [32]:

```
# Dataset overview

print("=" * 60)
print("Dataset Information")
print("=" * 60)

print(f"Rows: {raw_df.shape[0]}")
print(f"Columns: {raw_df.shape[1]}")

print("\nData Types")
display(raw_df.dtypes.value_counts().to_frame("Count"))

print("\nFirst 10 Columns")
display( pd.DataFrame(
        raw_df.columns[:10],
        columns=["Column Name"]     ))
```

```
============================================================
Dataset Information
============================================================
Rows: 1537
Columns: 145

Data Types
```

	Count
float64	141
object	3
int64	1

```
First 10 Columns
```

	Column Name
0	_id
1	Category
2	Topic
3	Attribute
4	City of Toronto
5	Agincourt North
6	Agincourt South-Malvern West
7	Alderwood
8	Annex
9	Banbury-Don Mills

In [34]:

```
# Missing Values Summary

missing_summary = raw_df.isnull().sum()

print("\nMissing Values Summary")
print(f"Columns with missing values: {(missing_summary > 0).sum()}")

print(
    f"Maximum missing values in a single column: "
    f"{missing_summary.max()}")

print(
    f"Total missing values: {missing_summary.sum()}")
```

```
Missing Values Summary
Columns with missing values: 140
Maximum missing values in a single column: 9
Total missing values: 1260
```

Interpretation

The initial inspection confirmed that the raw 2011 Toronto Neighbourhood Profiles dataset contains **1,537 socioeconomic indicators** across **145 variables**. The dataset follows the expected profile structure, where socioeconomic indicators are stored as rows and neighbourhoods are represented as columns.

The dataset consists primarily of numeric variables, with a small number of descriptive fields used for metadata. A limited number of missing values were identified, reflecting indicators that are not applicable to every neighbourhood. These will be addressed during feature extraction and data preparation.

Section-3 Research Variable Identification

Identify the socioeconomic indicators required for the temporal comparative analysis. The selected variables were chosen based on their relevance to neighbourhood socioeconomic vulnerability and their compatibility with the 2016 machine learning-ready dataset developed in Notebook 1.

This step establishes the analytical variables that will be extracted from the raw 2011 Toronto Neighbourhood Profiles and transformed into the machine learning-ready dataset.

In [9]:
```
# Research variables for temporal comparative analysis

research_variables = [
    "Population",
    "Participation Rate",
    "Employment Rate",
    "Unemployment Rate",
    "Bachelor",
    "No certificate",
    "Visible minority",
    "Renter",
    "Immigrant",
    "Lone-parent",
    "Low Income Measure After Tax"]

research_df = pd.DataFrame(
    {"Selected Research Variables": research_variables})

display(research_df)
```

	Selected Research Variables
0	Population
1	Participation Rate
2	Employment Rate
3	Unemployment Rate
4	Bachelor
5	No certificate
6	Visible minority
7	Renter
8	Immigrant
9	Lone-parent
10	Low Income Measure After Tax

Interpretation

The research variables were selected to support temporal comparison between the 2011 and 2016 Toronto Neighbourhood Profiles. These indicators represent the demographic, educational, labour market, housing, and socioeconomic characteristics identified as relevant to neighbourhood-level socioeconomic vulnerability.

The selected variables also maximize consistency with the 2016 machine learning-ready dataset, providing a common analytical framework for subsequent comparative analyses.

Section- 4 Locate Research Variables

Locate the selected socioeconomic indicators within the raw 2011 Toronto Neighbourhood Profiles dataset and record their corresponding row indices. This step establishes the mapping between the research variables and their locations in the raw dataset, providing a transparent and reproducible foundation for feature extraction.

In [11]:
```
# Locate research variables in the raw dataset

search_terms = [
    "Population, 2011",
    "Participation rate",
    "Employment rate",
    "Unemployment rate",
    "Bachelor",
    "No certificate",
    "Visible minority",
    "Renter",
```

```
    "Immigrant",
    "Lone-parent",
    "Low Income Measure After Tax"]

for term in search_terms:
    print("=" * 60)
    print(f"Search Term: {term}")
    print("=" * 60)

    display(
        raw_df[
            raw_df["Attribute"].str.contains(
                term,
                case=False,
                na=False            )
        ][["_id", "Category", "Topic", "Attribute"]]    )
```

```
============================================================
Search Term: Population, 2011
============================================================
```

	_id	Category	Topic	Attribute
0	1	Population	Population	Population, 2011

```
============================================================
Search Term: Participation rate
============================================================
```

	_id	Category	Topic	Attribute
1412	1413	Labour	Labour force status	Participation rate

```
============================================================
Search Term: Employment rate
============================================================
```

	_id	Category	Topic	Attribute
1413	1414	Labour	Labour force status	Employment rate
1414	1415	Labour	Labour force status	Unemployment rate

```
============================================================
Search Term: Unemployment rate
============================================================
```

	_id	Category	Topic	Attribute
1414	1415	Labour	Labour force status	Unemployment rate

```
============================================================
Search Term: Bachelor
============================================================
```

	_id	Category	Topic	Attribute
1372	1373	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1373	1374	Education	Highest certificate, diploma or degree	University certificate, diploma or degree ...
1374	1375	Education	Highest certificate, diploma or degree	Bachelor's degree
1375	1376	Education	Highest certificate, diploma or degree	University certificate, diploma or degre...
1382	1383	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1383	1384	Education	Highest certificate, diploma or degree	University certificate, diploma or degree ...
1384	1385	Education	Highest certificate, diploma or degree	Bachelor's degree
1385	1386	Education	Highest certificate, diploma or degree	University certificate, diploma or degre...

```
============================================================
Search Term: No certificate
============================================================
```

	_id	Category	Topic	Attribute
1367	1368	Education	Highest certificate, diploma or degree	No certificate, diploma or degree
1377	1378	Education	Highest certificate, diploma or degree	No certificate, diploma or degree

```
============================================================
Search Term: Visible minority
============================================================
```

	_id	Category	Topic	Attribute
995	996	Ethnocultural diversity	Visible minority population	Total population in private households by visi...
996	997	Ethnocultural diversity	Visible minority population	Total visible minority population
1007	1008	Ethnocultural diversity	Visible minority population	Visible minority, n.i.e.
1009	1010	Ethnocultural diversity	Visible minority population	Not a visible minority

```
============================================================
Search Term: Renter
============================================================
```

	_id	Category	Topic	Attribute
1305	1306	Housing	Housing tenure	Renter

```
============================================================
Search Term: Immigrant
============================================================
```

	_id	Category	Topic	Attribute
831	832	Immigration and citizenship	Immigrant status and period of immigration	Total population in private households by immi...
832	833	Immigration and citizenship	Immigrant status and period of immigration	Non-immigrants
833	834	Immigration and citizenship	Immigrant status and period of immigration	Immigrants
842	843	Immigration and citizenship	Age at immigration	Total immigrant population in private househol...
848	849	Immigration and citizenship	Immigrants by selected place of birth	Total population in private households by immi...
849	850	Immigration and citizenship	Immigrants by selected place of birth	Non-immigrants
852	853	Immigration and citizenship	Immigrants by selected place of birth	Immigrants
914	915	Immigration and citizenship	Recent immigrants by selected place of birth	Total recent immigrant population in private h...

```
============================================================
Search Term: Lone-parent
============================================================
```

	_id	Category	Topic	Attribute
153	154	Families, households and marital status	Household type	Lone-parent family households
159	160	Families, households and marital status	Household type	Lone-parent family households
193	194	Families, households and marital status	Family structure	Total lone-parent families by sex of parent ...
787	788	Income	Income of families	Lone-parent economic families

```
============================================================
Search Term: Low Income Measure After Tax
============================================================
```

_id	Category	Topic	Attribute

Section-5 Variable Selection and Justification

Review the candidate indicators identified in the raw dataset and select the most appropriate variable for each research construct. The selected indicators are based on conceptual relevance, consistency with the 2016 machine learning-ready dataset, and suitability for temporal comparative analysis.

In [12]:
```
# Final research variable selection

variable_selection = pd.DataFrame({

    "Research Variable": [

        "Population",
        "Participation_Rate",
        "Employment_Rate",
        "Unemployment_Rate",
        "Bachelors_Degree",
        "No_Diploma",
        "Visible_Minority",
        "Renters",
        "Immigrant_Population",
        "Lone_Parent_Families",
        "Low_Income_AfterTax_Pct"    ],

    "Selected Indicator": [

        "Population, 2011",
        "Participation rate",
        "Employment rate",
        "Unemployment rate",
        "Bachelor's degree",
        "No certificate, diploma or degree",
        "Total visible minority population",
        "Renter",
        "Immigrants",
        "Total lone-parent families",
        "Low-income measure after tax (LIM-AT)"    ],

    "Reason for Selection": [

        "Represents total neighbourhood population.",
        "Matches the 2016 labour force participation indicator.",
        "Matches the 2016 employment indicator.",
        "Matches the 2016 unemployment indicator.",
        "Represents the population whose highest educational attainment is a bachel
        "Represents residents without formal educational qualifications.",
        "Represents the total visible minority population at the neighbourhood leve
        "Represents renter households and aligns with the 2016 housing tenure indic
        "Represents the total immigrant population used in the 2016 analysis.",
        "Represents total lone-parent families within each neighbourhood.",
        "Represents neighbourhood low-income prevalence and serves as the dependent
```

```
display(variable_selection)
```

	Research Variable	Selected Indicator	Reason for Selection
0	Population	Population, 2011	Represents total neighbourhood population.
1	Participation_Rate	Participation rate	Matches the 2016 labour force participation in...
2	Employment_Rate	Employment rate	Matches the 2016 employment indicator.
3	Unemployment_Rate	Unemployment rate	Matches the 2016 unemployment indicator.
4	Bachelors_Degree	Bachelor's degree	Represents the population whose highest educat...
5	No_Diploma	No certificate, diploma or degree	Represents residents without formal educationa...
6	Visible_Minority	Total visible minority population	Represents the total visible minority populati...
7	Renters	Renter	Represents renter households and aligns with t...
8	Immigrant_Population	Immigrants	Represents the total immigrant population used...
9	Lone_Parent_Families	Total lone-parent families	Represents total lone-parent families within e...
10	Low_Income_AfterTax_Pct	Low-income measure after tax (LIM-AT)	Represents neighbourhood low-income prevalence...

In [13]:
```
raw_df[
    raw_df["Attribute"].str.contains(
        "low",
        case=False,
        na=False
    )
][["_id", "Category", "Topic", "Attribute"]]
```

Out[13]:

	_id	Category	Topic	Attribute
816	817	Income	Low income	In low income in 2010 based on after-tax low-i...
821	822	Income	Low income	Prevalence of low income in 2010 based on afte...
1372	1373	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1382	1383	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...

Interpretation

The candidate indicators identified in the raw 2011 Toronto Neighbourhood Profiles dataset were systematically reviewed to select the most appropriate variable for each research construct. The selection process was guided by conceptual relevance, data availability, and consistency with the 2016 machine learning-ready dataset.

Where multiple candidate indicators were available, the indicator that most closely represented the corresponding socioeconomic concept was selected. This approach ensured that the final feature set provides a transparent, reproducible, and methodologically consistent foundation for temporal comparative analysis between the 2011 and 2016 datasets.

Section-6. Final Feature Mapping

Create a final feature mapping that links each selected research variable to its corresponding indicator and source row within the raw 2011 Toronto Neighbourhood Profiles dataset. This mapping provides a transparent and reproducible reference for extracting the selected variables and constructing the machine learning-ready dataset.

In [15]:

```
# Final feature mapping

mapping_df = pd.DataFrame({

    "Research Variable": [

        "Population",
        "Participation_Rate",
        "Employment_Rate",
        "Unemployment_Rate",
        "Bachelors_Degree",
        "No_Diploma",
        "Visible_Minority",
        "Renters",
        "Immigrant_Population",
        "Lone_Parent_Families",
```

```
        "Low_Income_AfterTax_Pct"    ],

    "Selected Indicator": [

        "Population, 2011",
        "Participation rate",
        "Employment rate",
        "Unemployment rate",
        "Bachelor's degree",
        "No certificate, diploma or degree",
        "Total visible minority population",
        "Renter",
        "Immigrants",
        "Total lone-parent families",
        "Prevalence of low income in 2010 based on after-tax low-income measure (LI

    "Source Row": [

        0,
        1412,
        1413,
        1414,
        1384,
        1367,
        996,
        1305,
        833,
        193,
        821    ]})

display(mapping_df)
```

	Research Variable	Selected Indicator	Source Row
0	Population	Population, 2011	0
1	Participation_Rate	Participation rate	1412
2	Employment_Rate	Employment rate	1413
3	Unemployment_Rate	Unemployment rate	1414
4	Bachelors_Degree	Bachelor's degree	1384
5	No_Diploma	No certificate, diploma or degree	1367
6	Visible_Minority	Total visible minority population	996
7	Renters	Renter	1305
8	Immigrant_Population	Immigrants	833
9	Lone_Parent_Families	Total lone-parent families	193
10	Low_Income_AfterTax_Pct	Prevalence of low income in 2010 based on afte...	821

Interpretation

The final feature mapping establishes a direct relationship between each selected research variable, its corresponding socioeconomic indicator, and its source row within the raw 2011 Toronto Neighbourhood Profiles dataset.

The mapped variables include the independent variables used to explain neighbourhood socioeconomic conditions and the dependent variable, **Low_Income_AfterTax_Pct**, which represents neighbourhood-level low-income prevalence. Documenting the selected indicator and source row for each variable ensures that the feature extraction process is transparent, reproducible, and methodologically consistent for constructing the 2011 machine learning-ready dataset.

Section-7 Feature Extraction

Extract the selected socioeconomic indicators from the raw 2011 Toronto Neighbourhood Profiles dataset using the final feature mapping. This step creates a subset containing only the research variables required for constructing the machine learning-ready dataset.

In [22]:

```
# Extract selected variables

selected_df = raw_df.iloc[
    list(feature_mapping.values())].copy()

print("=" * 60)
print("Feature Extraction Summary")
print("=" * 60)

print(f"Selected Variables: {selected_df.shape[0]}")
print(f"Dataset Columns: {selected_df.shape[1]}")

display(selected_df[["_id", "Category", "Topic", "Attribute"] ])
```

```
============================================================
Feature Extraction Summary
============================================================
Selected Variables: 11
Dataset Columns: 145
```

	_id	Category	Topic	Attribute
0	1	Population	Population	Population, 2011
1412	1413	Labour	Labour force status	Participation rate
1413	1414	Labour	Labour force status	Employment rate
1414	1415	Labour	Labour force status	Unemployment rate
1384	1385	Education	Highest certificate, diploma or degree	Bachelor's degree
1367	1368	Education	Highest certificate, diploma or degree	No certificate, diploma or degree
996	997	Ethnocultural diversity	Visible minority population	Total visible minority population
1305	1306	Housing	Housing tenure	Renter
833	834	Immigration and citizenship	Immigrant status and period of immigration	Immigrants
193	194	Families, households and marital status	Family structure	Total lone-parent families by sex of parent ...
821	822	Income	Low income	Prevalence of low income in 2010 based on afte...

Interpretation

The selected socioeconomic indicators were successfully extracted from the raw 2011 Toronto Neighbourhood Profiles dataset using the final feature mapping. The extracted subset contains **11 research variables** and preserves the original profile structure, where socioeconomic indicators remain as rows and neighbourhoods as columns.

Validation of the extracted records confirmed that each research variable corresponds to the intended socioeconomic indicator identified during the variable selection process. This verified subset provides the foundation for transforming the data into a neighbourhood-level machine learning-ready dataset for subsequent temporal comparative analysis.

Section-8 Trasform data structure

Transform the extracted socioeconomic indicators from the original profile format into a neighbourhood-level analytical dataset. This process converts socioeconomic indicators from rows to columns, creating a machine learning-ready structure where each row represents a neighbourhood and each column represents a research variable.

In [23]:
```
# Transform the dataset into neighbourhood-level observations

ml_ready_df = (
```

```
    selected_df
    .drop(columns=["_id", "Category", "Topic", "Attribute"])
    .transpose())

ml_ready_df.columns = list(feature_mapping.keys())

ml_ready_df.index.name = "Neighbourhood"

ml_ready_df = ml_ready_df.reset_index()

print("=" * 60)
print("Dataset Transformation Summary")
print("=" * 60)

print(f"Rows (Neighbourhoods): {ml_ready_df.shape[0]}")
print(f"Columns (Variables): {ml_ready_df.shape[1]}")

display(ml_ready_df.head())
```

```
============================================================
Dataset Transformation Summary
============================================================
Rows (Neighbourhoods): 141
Columns (Variables): 12
```

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	E
0	City of Toronto	2615060.0	64.3	58.3	9.3	
1	Agincourt North	30279.0	57.9	51.3	11.3	
2	Agincourt South-Malvern West	21988.0	59.3	53.0	10.7	
3	Alderwood	11904.0	66.0	61.1	7.4	
4	Annex	29177.0	71.8	66.6	7.3	

Section-9 Remove the City of Toronto Aggregate Row

Remove the City of Toronto aggregate record from the transformed dataset. The aggregate represents a city-wide summary rather than an individual neighbourhood and is therefore excluded to ensure that all observations correspond to comparable neighbourhood-level units for temporal comparative analysis.

In [28]:
```
# Remove the City of Toronto aggregate row

ml_ready_df = (
    ml_ready_df[
        ml_ready_df["Neighbourhood"] != "City of Toronto" ].reset_index(drop=True))

print("City of Toronto Present:",
    "City of Toronto" in ml_ready_df["Neighbourhood"].values)
```

```
print("=" * 60)
print("Neighbourhood Dataset Summary")
print("=" * 60)

print(f"Neighbourhoods: {ml_ready_df.shape[0]}")
print(f"Research Variables: {ml_ready_df.shape[1] - 1}")
print("Identifier Column: 1")
print(f"Total Columns: {ml_ready_df.shape[1]}")

display(ml_ready_df.head())
```

```
City of Toronto Present: False
============================================================
Neighbourhood Dataset Summary
============================================================
Neighbourhoods: 140
Research Variables: 11
Identifier Column: 1
Total Columns: 12
```

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	E
0	Agincourt North	30279.0	57.9	51.3	11.3	
1	Agincourt South-Malvern West	21988.0	59.3	53.0	10.7	
2	Alderwood	11904.0	66.0	61.1	7.4	
3	Annex	29177.0	71.8	66.6	7.3	
4	Banbury-Don Mills	26918.0	61.3	57.0	7.1	

Interpretation

The City of Toronto aggregate record was successfully removed from the machine learning-ready dataset because it represents a city-wide summary rather than an individual neighbourhood. Including this aggregate observation could bias descriptive statistics and subsequent analyses, as it is not directly comparable to neighbourhood-level records.

The resulting dataset contains **140 neighbourhood-level observations**, **11 research variables** (10 independent variables and 1 dependent variable), and a single identifier column (`Neighbourhood`). This finalized structure provides a consistent analytical dataset for temporal comparison with the 2016 machine learning-ready dataset and for subsequent statistical analyses and predictive modelling.

Section-10 Data Quality Assessment

Evaluate the quality of the machine learning-ready dataset by examining its structure, data types, missing values, and duplicate records. This assessment verifies that the transformed dataset is complete, internally consistent, and suitable for subsequent temporal comparative analysis.

In [30]:
```
# Data quality assessment

print("=" * 60)
print("Data Quality Summary")
print("=" * 60)

print(f"Dataset Shape: {ml_ready_df.shape}")

print("\nData Types")

dtype_summary = (
    ml_ready_df.dtypes
    .value_counts()
    .rename_axis("Data Type")
    .reset_index(name="Count"))

display(dtype_summary)

print("\nMissing Values")

missing_values = ml_ready_df.isnull().sum().sum()
print(f"Total Missing Values: {missing_values}")

print("\nDuplicate Records")

duplicates = ml_ready_df.duplicated().sum()
print(f"Duplicate Records: {duplicates}")
```

```
============================================================
Data Quality Summary
============================================================
Dataset Shape: (140, 12)

Data Types
```

	Data Type	Count
0	float64	11
1	object	1

```
Missing Values
Total Missing Values: 0

Duplicate Records
Duplicate Records: 0
```

Interpration

The data quality assessment confirmed that the transformed dataset is complete, internally consistent, and suitable for subsequent analysis. The final machine learning-ready dataset contains **140 neighbourhood-level observations**, **11 numeric research variables**, and a single identifier column (`Neighbourhood`).

The validation identified **no missing values** and **no duplicate records**, indicating that the data preparation and transformation processes were successfully completed. The resulting dataset is fully prepared for temporal comparative analysis and can be used directly in the statistical analyses and modelling presented in Notebook 5.

Section-11. Export the Machine Learning-Ready Dataset

Export the validated 2011 machine learning-ready dataset to a CSV file for use in subsequent temporal comparative analysis. This step preserves the final analytical dataset and provides a standardized input for Notebook 5.

In [38]:
```
# Export the machine learning-ready dataset

output_path = "../../data/processed/neighbourhood_profiles_2011_ml_ready.csv"

ml_ready_df.to_csv(
    output_path,
    index=False)

print("=" * 60)
print("Dataset Export Summary")
print("=" * 60)

print("Export completed successfully.")
print(f"Output File: {output_path}")
print(f"Dataset Shape: {ml_ready_df.shape}")
```

```
============================================================
Dataset Export Summary
============================================================
Export completed successfully.
Output File: ../../data/processed/neighbourhood_profiles_2011_ml_ready.csv
Dataset Shape: (140, 12)
```

12. Conclusion

This notebook prepared a machine learning-ready dataset from the 2011 Toronto Neighbourhood Profiles to support temporal comparative analysis with the 2016 dataset developed in Notebook 1.

A systematic feature engineering workflow was implemented to identify, justify, extract, transform, and validate the selected socioeconomic indicators. The original profile-style dataset was converted into a neighbourhood-level analytical dataset, and the City of Toronto aggregate record was removed to ensure that all observations represented comparable neighbourhood-level units.

The final dataset contains **140 neighbourhoods**, **11 research variables** (10 independent variables and 1 dependent variable), and **one identifier column** (`Neighbourhood`). Data quality assessment confirmed that the dataset contains no missing values or duplicate records, indicating that it is complete, internally consistent, and suitable for subsequent temporal comparative analysis.

The exported machine learning-ready dataset provides a transparent, reproducible, and methodologically consistent foundation for the comparative analyses presented in Notebook 5.